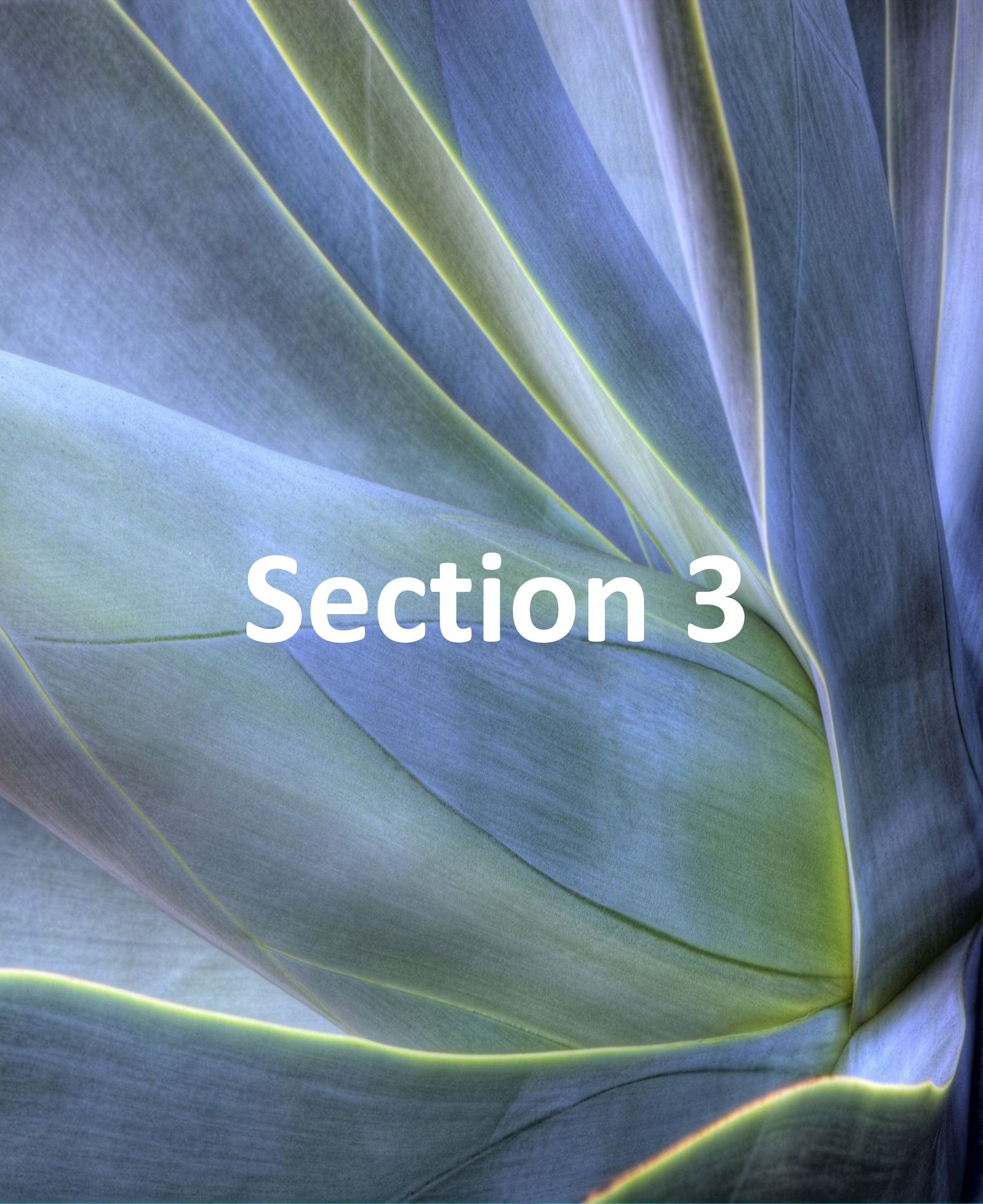
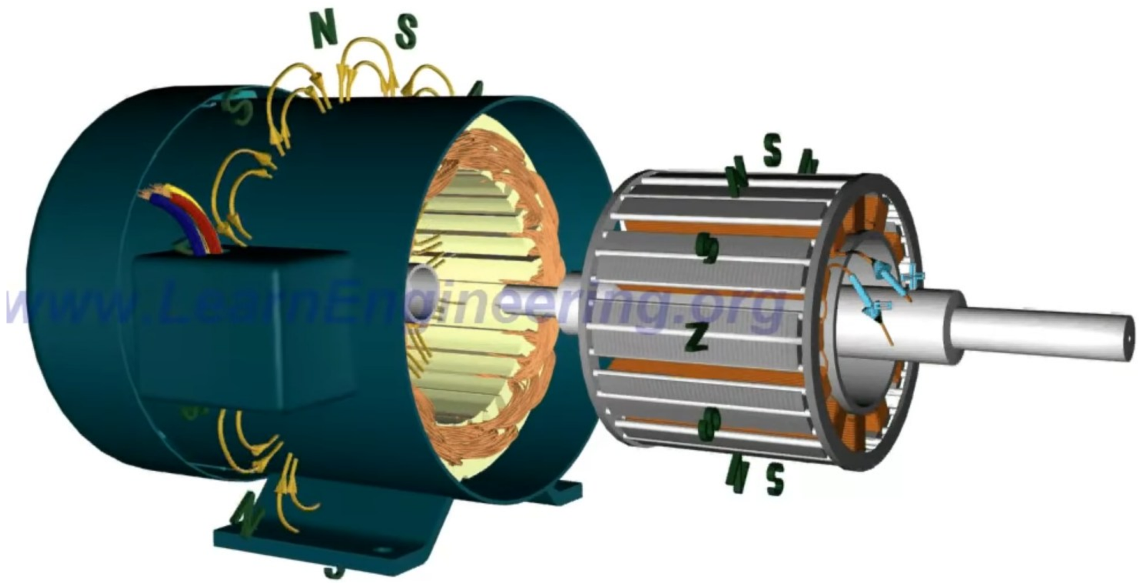


Section 3

Synchronous motor

1. Introduction Synchronous Motor

- A **synchronous motor** is an **AC motor** in which the **rotor rotates at the same speed as the rotating magnetic field** of the stator — hence the term "*synchronous*".
- This means there is **no slip** between the speed of the stator magnetic field and the rotor speed.
- **Synchronous motors are widely used in:**
  - ➢ Large-scale power generation (as synchronous generators/alternators)
  - ➢ Industrial drives requiring constant speed:
    - Compressors
    - Blowers and fans
    - Rolling mills
    - Pumps
    - Textile and paper mills



2. Synchronous Motor vs Asynchronous Motor

Items	Synchronous Motor	Induction Motor
Speed	Constant (no slip)	Varies with load (slip present)
Starting	Not self-starting	Self-starting
Power factor	Can be lagging, unity, or leading	Always lagging
Efficiency	High	Moderate
Excitation	Requires DC	No excitation needed

3. Construction

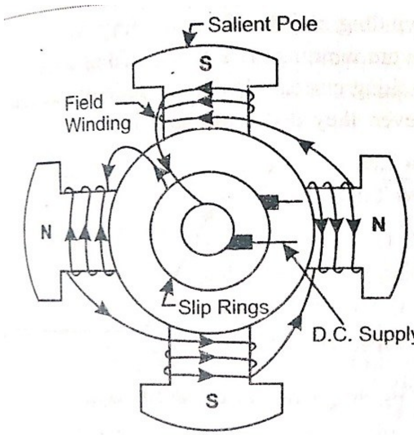
A synchronous motor mainly has three parts:

- **Stator:**
  - ➢ It is the stationary part that carries a three-phase winding.
  - ➢ When the stator winding is energized by a three-phase AC supply, it produces a rotating magnetic field.
  - ➢ The stator construction is similar to that of an induction motor.

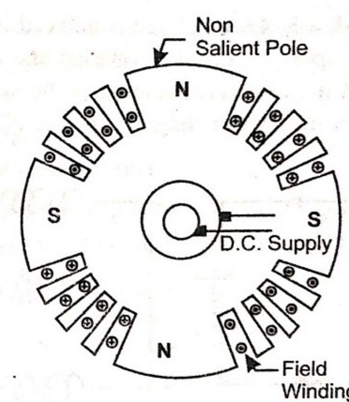
- **Rotor:**

Two types of rotor constructions are common:

- ➢ Salient Pole Rotor
  - Used for low-speed applications (below 600 rpm).
  - Poles are projected (salient).
  - Examples: Hydroelectric generators, large synchronous motors.
- ➢ Non-salient (Cylindrical) Rotor
  - Used for high-speed applications (above 1500 rpm).
  - The rotor is smooth and cylindrical, with field windings embedded in slots.
  - Examples: Turbo alternators, high-speed compressors.



Salient Pole Construction

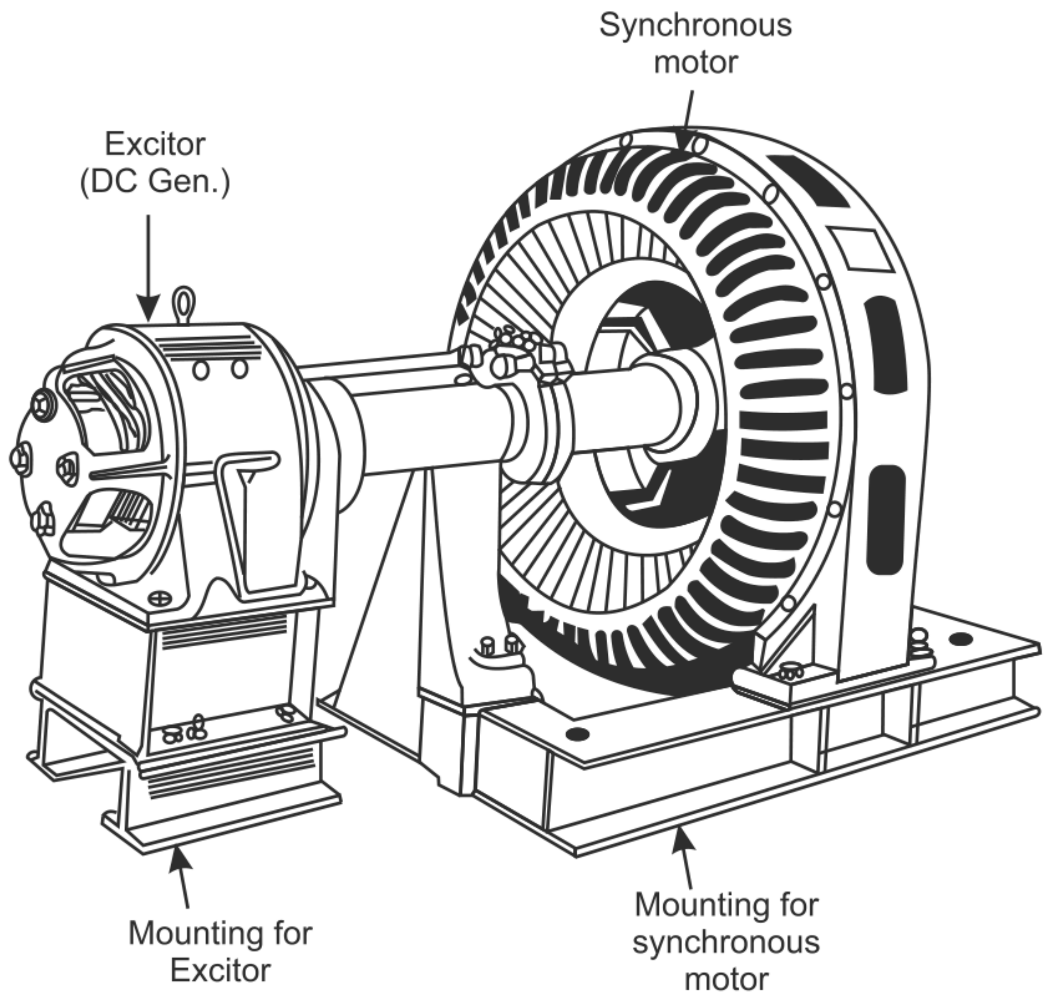


Non-salient Pole Construction

3. Construction

A synchronous motor mainly has three parts:

- **Excitation System:** Provides DC current to the rotor winding to produce the magnetic field.



4. Working Principle

A **synchronous motor** operates on the principle of magnetic locking (synchronization) between the rotating magnetic field (RMF) produced by the stator and the magnetic field of the rotor (created by DC excitation).

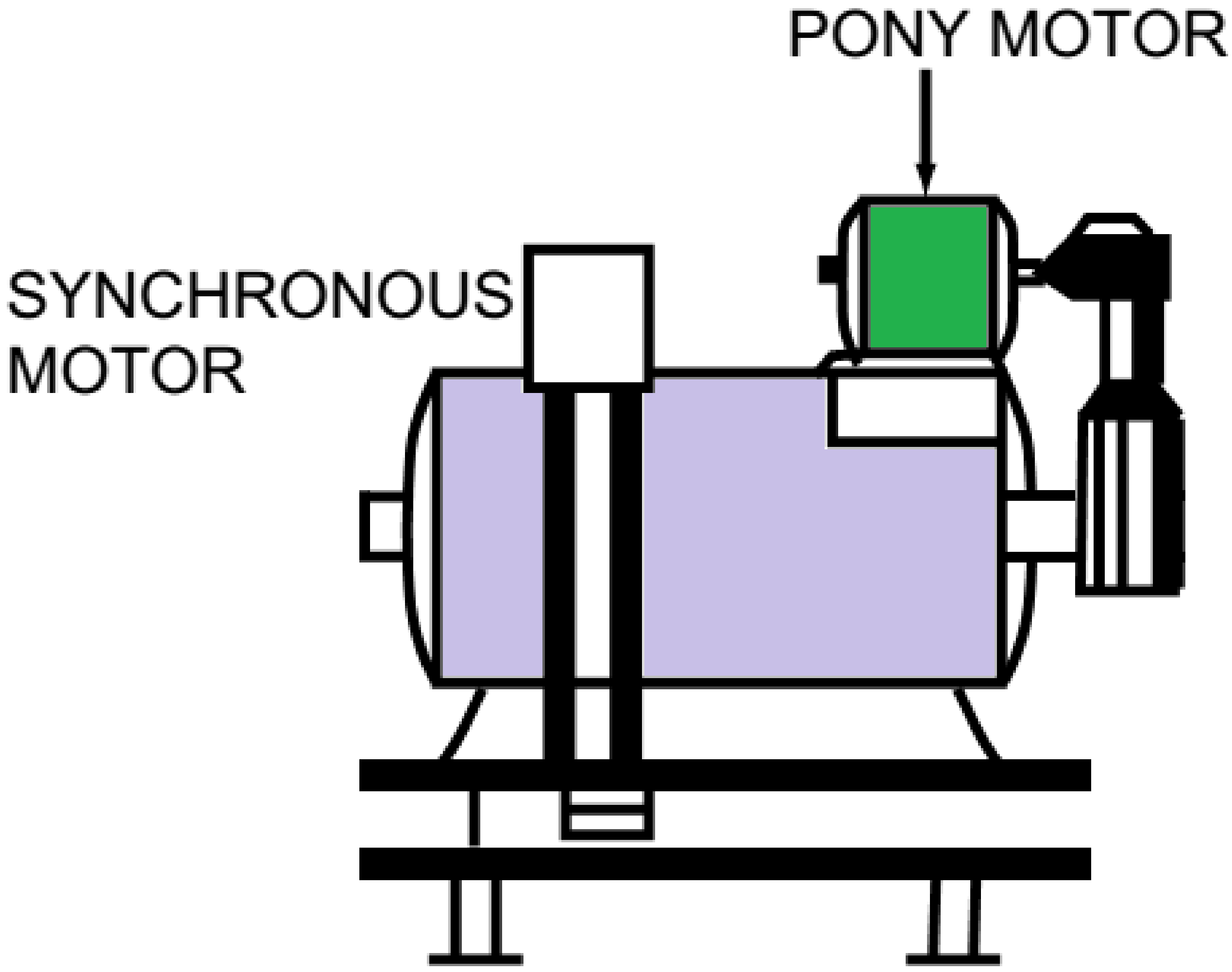
1. **Production of Rotating Magnetic Field (RMF) in the Stator:**
   - When a three-phase AC supply is given to the stator winding, it produces a rotating magnetic field (RMF).
   - The RMF rotates at a constant speed, called the synchronous speed (Ns).
2. **Production of Magnetic Field in the Rotor:**
   - A DC current is supplied to the rotor field winding through slip rings or via a brushless excitation system.
   - This produces a steady magnetic field — i.e., the rotor becomes an electromagnet with fixed North (N) and South (S) poles.
3. **So now we have two magnetic fields :**
   - Rotating magnetic field from the stator (RMF)
   - Stationary magnetic field (from the rotor)

4. Working Principle

4. Magnetic Interaction and Synchronization

- If the rotor is stationary, the rotating stator field sweeps past the rotor poles very quickly (at synchronous speed), changing the direction of magnetic forces many times per second.
- Because of rotor inertia, it cannot follow these rapid rotating. This is why the synchronous motor is not self-starting.
- To achieve rotation, the rotor must first be brought close to synchronous speed using:
  - Damper windings acting as an induction motor during start-up.
  - A separate starting motor (pony motor), or
- When the rotor reaches near-synchronous speed:
  - The opposite poles attract (N to S, S to N).
  - The two fields lock magnetically — this is called synchronism or magnetic locking.
  - Once magnetic locking occurs, the rotor poles are "caught" by the rotating stator field. From then on, both fields rotate together at the same speed.

Pony Motor



4. Working Principle

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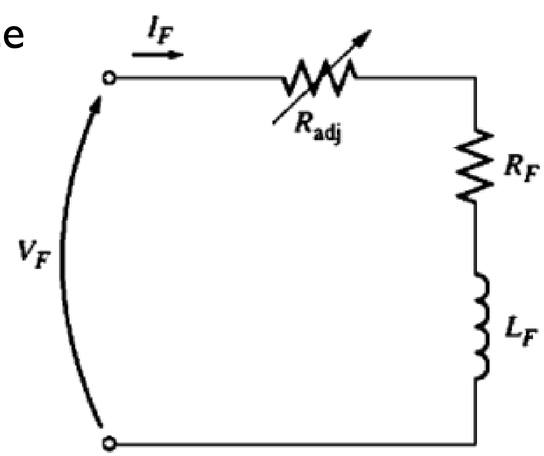
5. The Equivalent Circuit of a Synchronous Motor

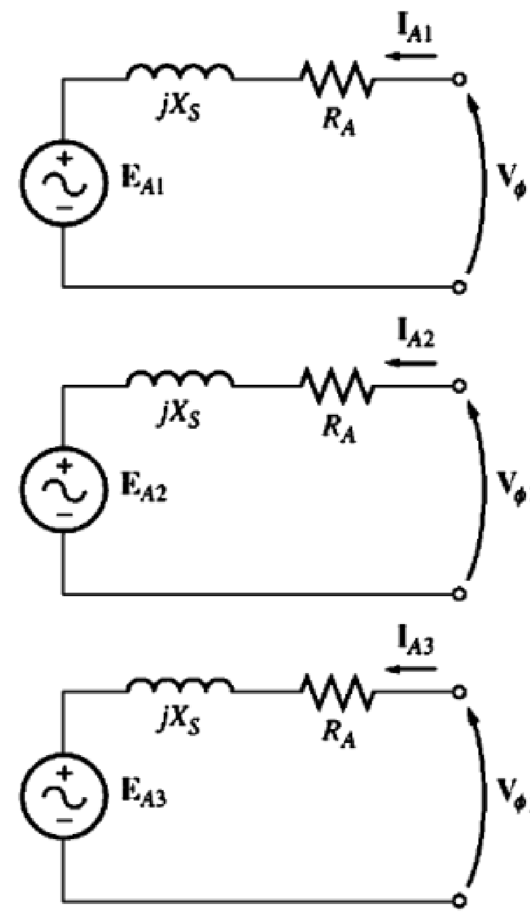
This diagram shows the **equivalent circuit of a synchronous motor**, broken into **field excitation (DC side)** and **armature (AC side)** representations.

- **Left Side — DC Field Circuit (Excitation System)**

  This represents the rotor field winding, which is supplied with DC current to produce a magnetic field.

  V_F: Field voltage (DC excitation voltage)
  I_F : Field current
  R_{Adj}: Adjustable resistor (controls I_F)
  R_F: Field winding resistance
  L_F: Field winding inductance





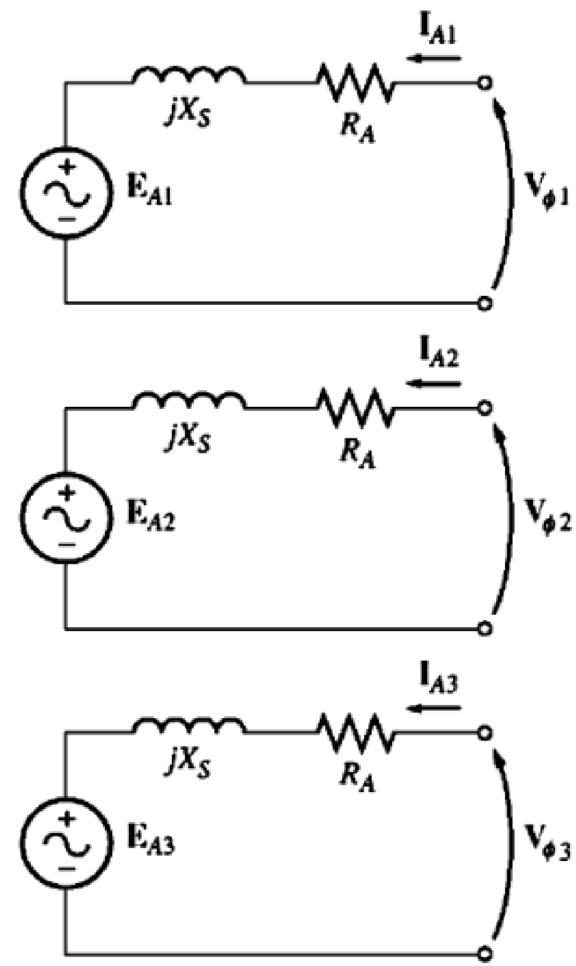
5. The Equivalent Circuit of a Synchronous Motor

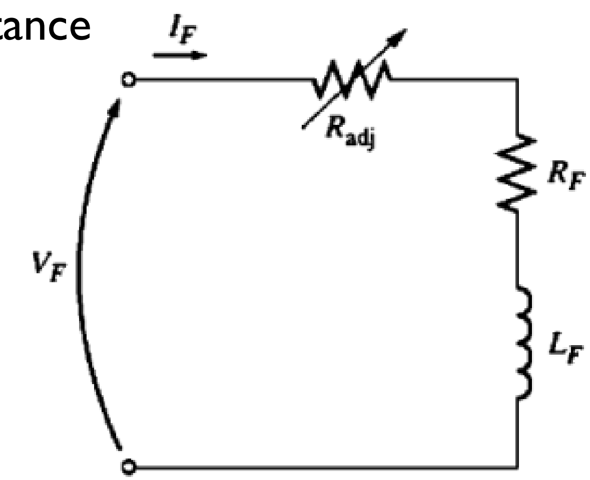
- **Right Side — Armature Circuit (AC Side)**

It is the **stator (armature) equivalent circuit** of the synchronous motor.

$$V_\phi = E_A + jX_S I_A + R_A I_A$$

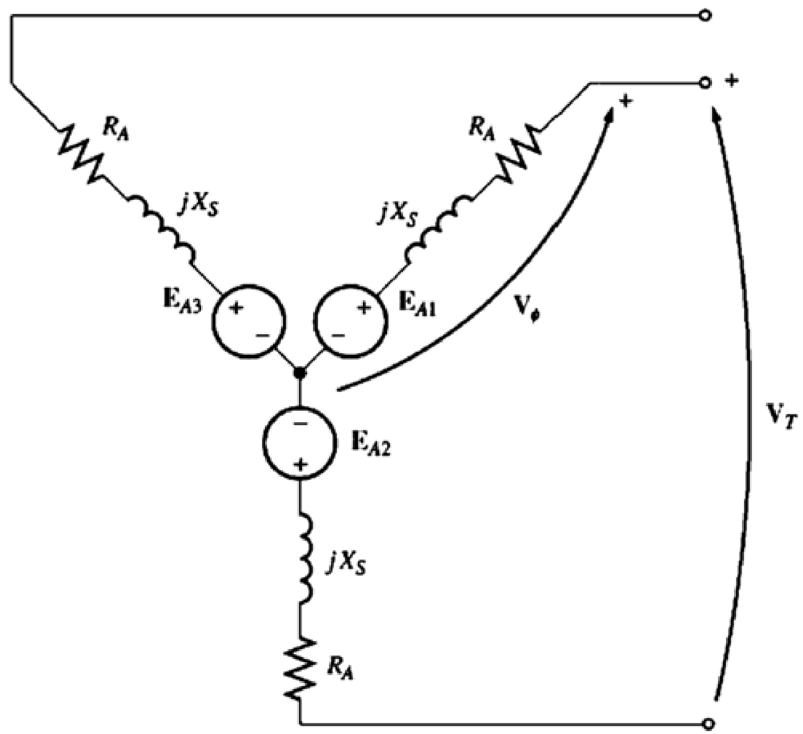
V_ϕ: Per-phase applied voltage (stator terminal voltage)
E_A: Internal generated EMF (due to rotor field excitation)
I_A: Armature current
R_A: Armature resistance
X_S: Synchronous reactance



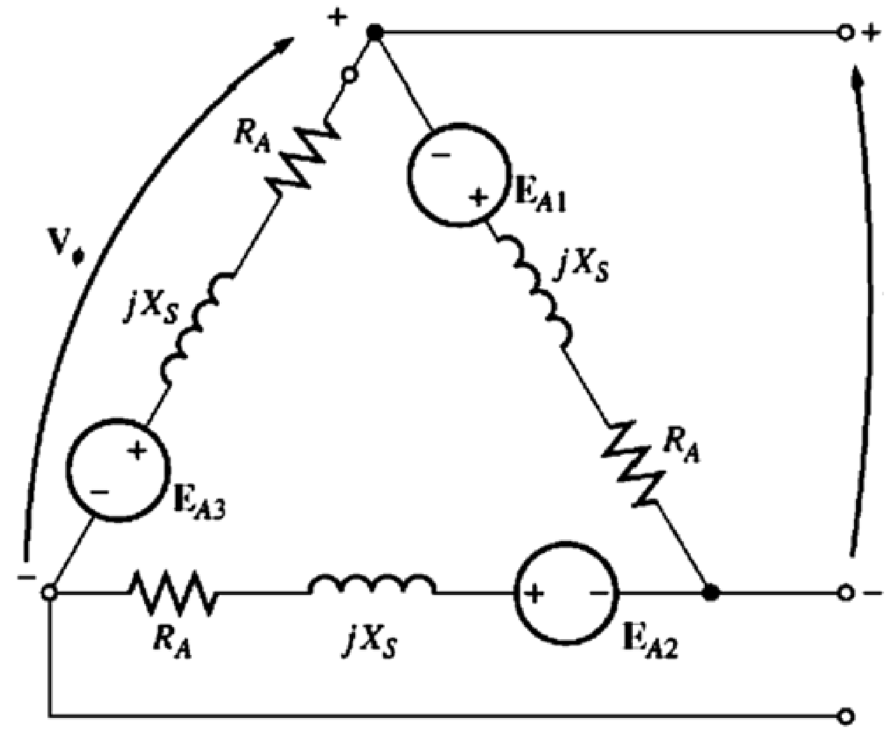


6. Y/Δ-connection

Y-connected



Δ-connected



Applications:

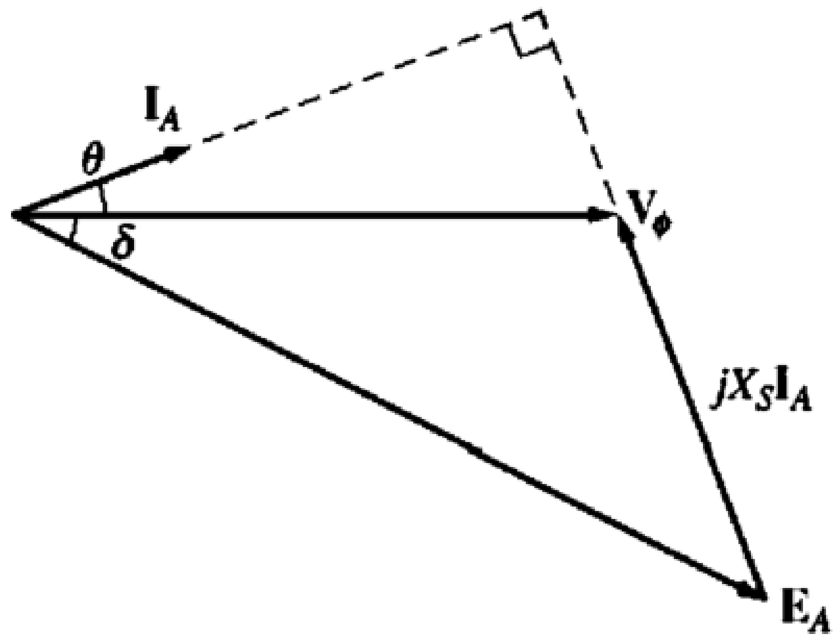
- For high-voltage, low-current operation
- Easier neutral grounding
- Smooth starting with reduced current
- Common in large industrial synchronous motors

Applications:

- For low-voltage, high-current operation
- More robust against unbalanced loads
- Used where no neutral is needed

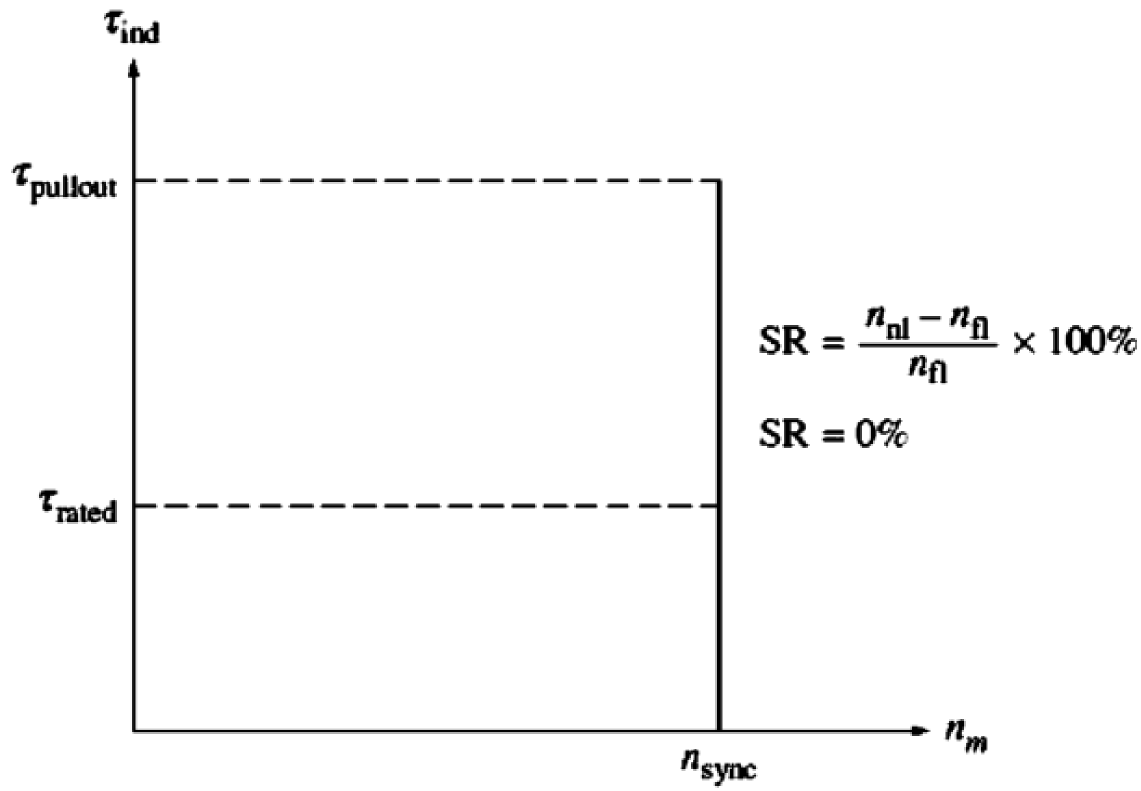
7. Phasor Diagram

$$V_\phi = E_A + jX_S I_A + R_A I_A$$



- $\mathbf{I_A}$ such that **I** leads **V** if the machine is overexcited or lags **V** if underexcited.
- $\mathbf{E_A}$ to lag behind V by the torque angle **δ.**

8. Torque-Speed Characteristic Curve



- Torque equation is: $$\tau_{ind} = \frac{3V_\phi E_A \sin\delta}{\omega_m X_S}$$

- The maximum torque : $$\tau_{\max} = \frac{3V_\phi E_A}{\omega_m X_S}$$

8. Torque-Speed Characteristic Curve

- **Constant Speed Operation:** This speed does not change from no load up to the motor's maximum torque capability.

- **Zero Speed Regulation:** Because the speed is constant regardless of the load, the motor has a speed regulation of 0%.

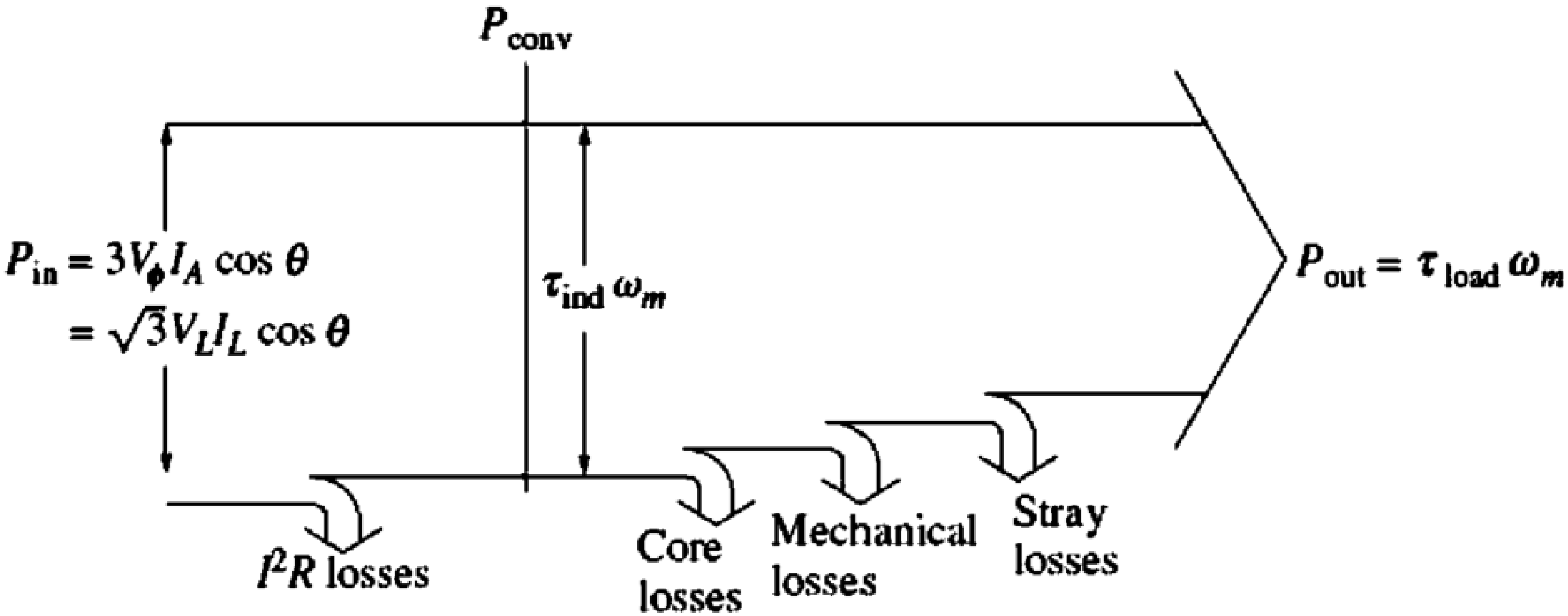
- **Pullout Torque:** The maximum torque the motor can produce is called the **pullout torque**. If the load torque exceeds this value, the motor loses synchronization.

- **Torque Equations:** The induced torque (τ_{ind}) depends on the sine of the angle (δ) between the rotor and stator magnetic fields. The maximum torque (τ_{max}) occurs when this angle is 90°.

- **Slipping Poles:** When the shaft load exceeds the pullout torque, the rotor can no longer stay in sync with the stator's rotating magnetic field. It starts to slip, this condition is known as **slipping poles**.

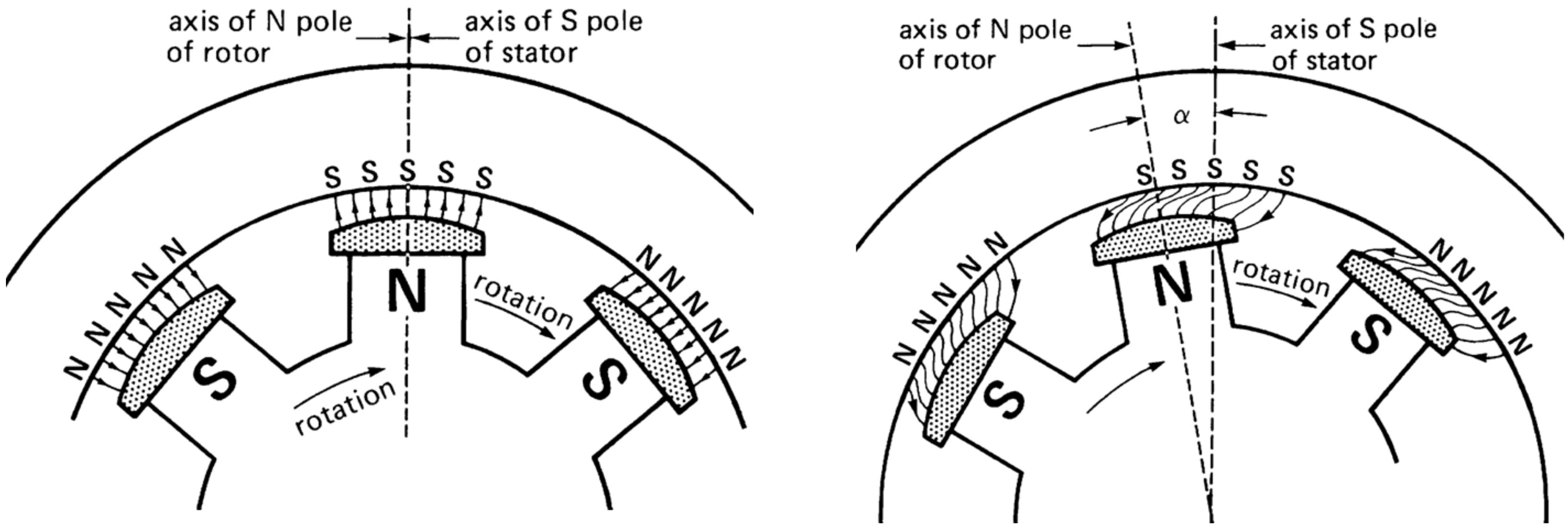
9. Power diagram

This diagram is a standard representation for rotating AC machines and applies perfectly to the synchronous motor.



1. Electrical power enters the stator.
2. Some power is immediately lost in the stator as heat RI^2.
3. The remaining power (P_{conv}) cross the air gap and is converted to mechanical power.
4. Part of this mechanical power is used to overcome friction, windage, and other stray losses.
5. The final remaining power (P_{out}) is delivered to the load.

10. The Effect of Load Changes on a Synchronous Motor

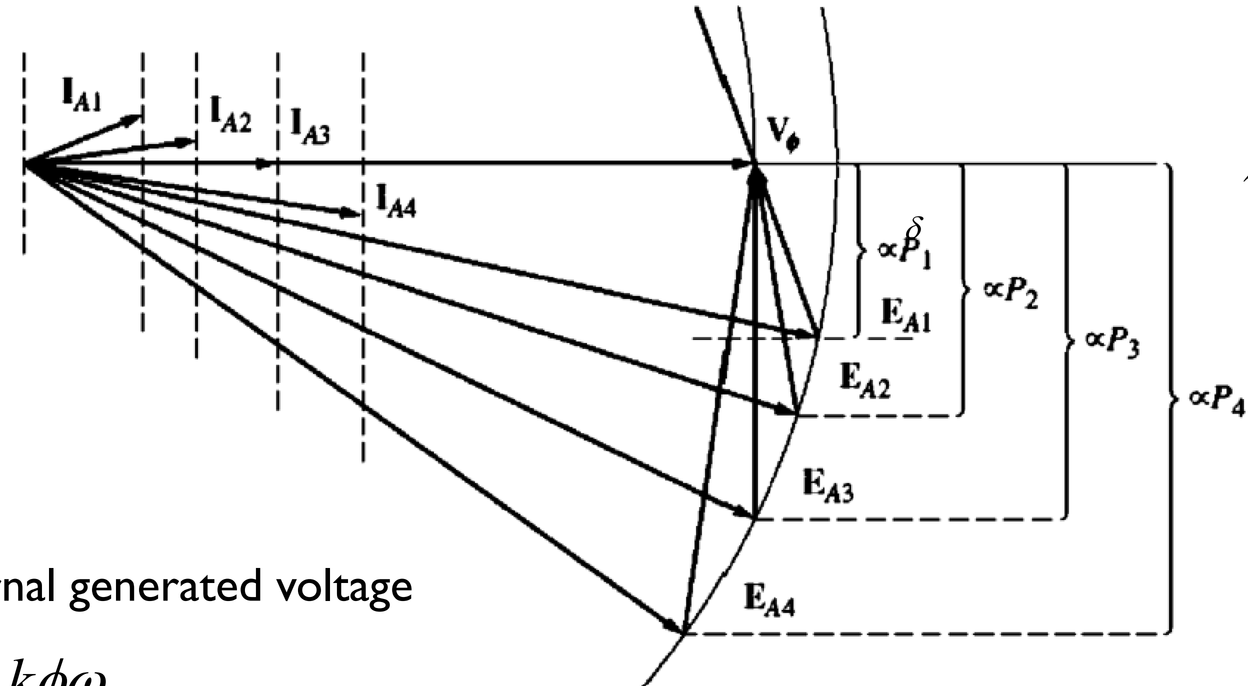


- At no-load conditions, the rotor poles are directly opposite the stator poles and their axes coincide.
- As mechanical load is applied, the rotor poles fall slightly behind the stator poles, but continues to turn at synchronous speed.
- Greater torque is developed with increase separation angle.
- There is a limit when the mechanical load exceeds the pull-out torque; the motor will stall and come to a halt.

11. The Effect of Load Changes on a Synchronous Motor

What happens when the load is changed on a synchronous motor?

- If the load on the shaft of the motor is increased, the rotor will initially slow down. As it does, the torque angle δ becomes larger, and the induced torque increases.



$$\tau_{ind} = \frac{3V_\phi E_A \sin\delta}{\omega_m X_S}$$

$$P = \frac{3V_\phi E_A \sin\delta}{X_S}$$

Internal generated voltage

$$E_A = k\phi\omega_m$$

k = machine constant,

Φ = rotor flux per pole (depends on DC excitation current),

ω_m = angular speed of rotation (synchronous speed).

Example 1

A 3-phase synchronous motor (Y-connected) is fed from a 415V (line) balanced supply. The motor's synchronous reactance per phase is $X_s = 2.0\,\Omega$. The internal generated (induced) voltage per phase is $E_A = 230\,V$(magnitude). Neglect stator resistance, core losses, rotational/mechanical losses.

1. Compute the maximum power the motor can deliver (electrical air-gap power) $P_{\max}$?
2. If the motor must deliver $P = 50\,kW$(mechanical load reflected to air-gap), find the torque (load) angle δ?
3. Find the per-phase armature current magnitude and its phase angle?

Example 2

A 208-V, Δ-connected, 60-Hz synchronous machine has a synchronous reactance of 2.5 Ω and a negligible armature resistance. Its friction and windage losses are 1.5 kW, and its core losses are 1.0 kW. Initially, the shaft is supplying a 15-hp load, and the motor's power factor is 0.80 leading.

1. Draw the phasor diagram of this motor, and find the values of I_A , I_L and $E_{A.}$

2. Assume that the shaft load is now increased to 30 hp. What is the new torque angle? Draw the behavior of the phasor diagram in response to this change. Find I_A and I_L after the load change?

12. The Effect of Field Current Changes

What effect does a change in field current have on a synchronous motor?

- The field current controls the magnitude of the internal generated voltage (E_A).
- It does not affect the real power supplied by the motor. Power supplied by the motor changes only when the shaft load torque changes.